2D Plane Stress Analysis of Plate with Circular Hole

FEM Solver Development and ANSYS Validation

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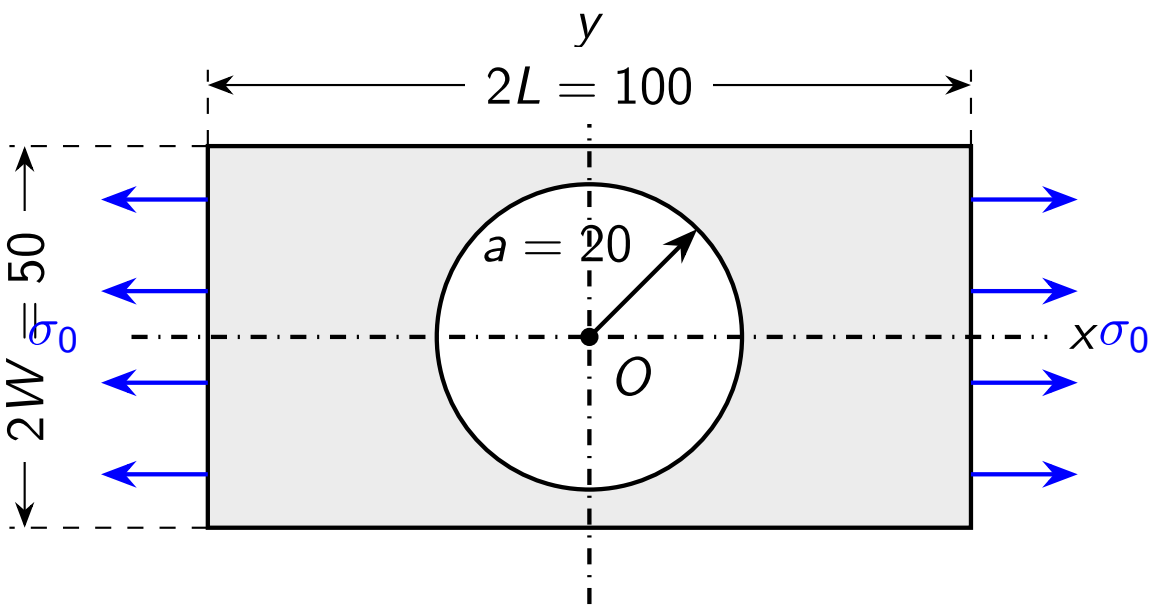
Problem Statement

- Perform 2D plane stress analysis of a rectangular plate with a circular hole under uniaxial tension.
- Develop a custom Finite Element Method (FEM) solver in MATLAB to compute:
  - X - Displacement
  - Stress distribution
  - Strain energy
- Validate the accuracy of the developed solver by comparing results with ANSYS simulations.
- Assess convergence behavior and compare with the analytical stress concentration factor ($K_t \approx 3$).

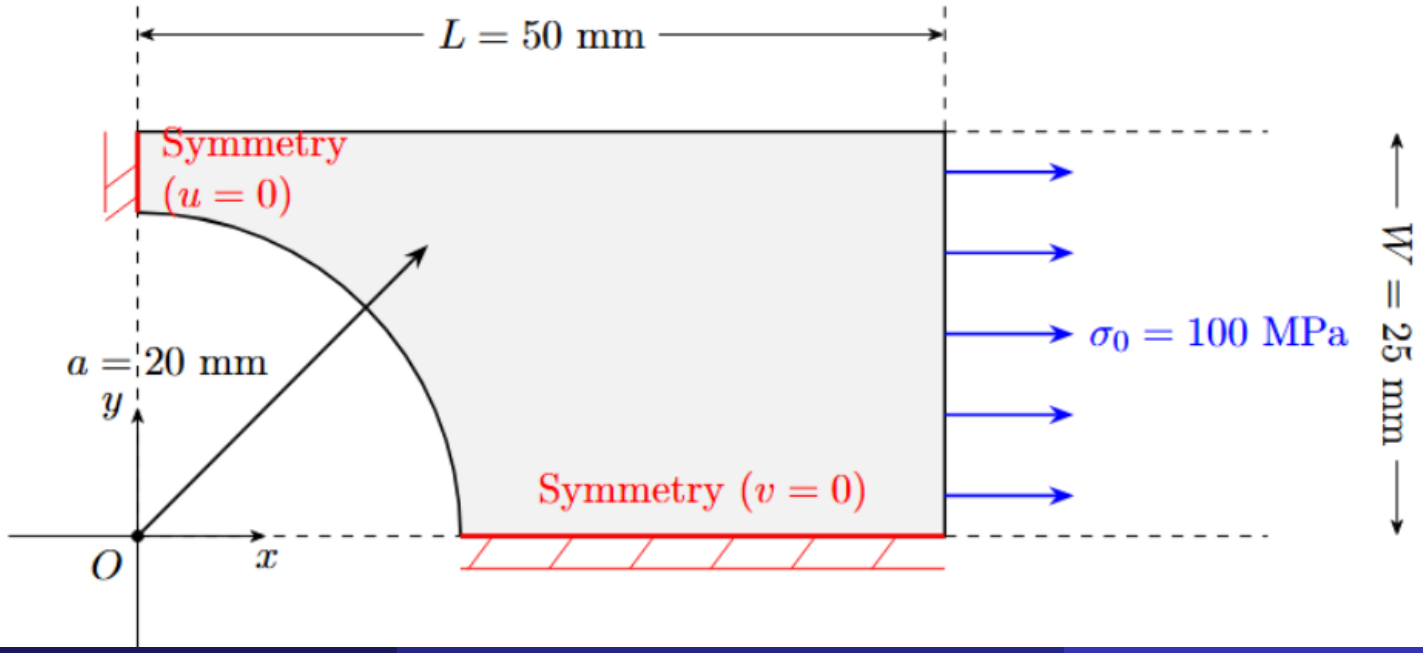
Geometry and Loading

- **Full Plate Size:**
  $2L \times 2W = 100 \times 50$ mm
- **Hole Radius:**
  $r = a = 20$ mm
- **Model Used:**
  Quarter-symmetry utilized for solver efficiency
  (50×25 mm domain)
- **Loading:**
  Uniaxial Tension
  $\sigma_0 = 100$ MPa



Boundary Conditions

- **Left edge ($x = 0$):**
  Roller support, $u_x = 0$ (Y-axis sym.)
- **Bottom edge ($y = 0$):**
  Roller support, $u_y = 0$ (X-axis sym.)
- **Right edge ($x = L$):**
  Uniform tension, $\sigma_0 = 100$ MPa
- **Hole boundary ($r = a$):**
  Traction-free surface



Material Properties

- Young's Modulus: $E = 210000$ MPa
- Poisson's Ratio: $\nu = 0.3$
- Plane stress condition assumed

FEM Formulation and Element Discretization

1. Isoparametric Q4 Elements

- Bilinear shape functions map natural (ξ, η) to physical (x, y) domains:

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta)$$

- The Jacobian matrix (J) is required to compute physical spatial derivatives:

$$\begin{Bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \end{Bmatrix} = J^{-1} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{Bmatrix}$$

2. Element Stiffness Matrix

- Derived via Principle of Virtual Work:

$$K_e = t \int_{-1}^{1} \int_{-1}^{1} B^T D B \det(J) \, d\xi \, d\eta$$

- Evaluated numerically using 2×2 **Gauss-Legendre Quadrature**:

$$K_e \approx t \sum_{i=1}^{2} \sum_{j=1}^{2} \left[B^T D B \det(J) \right]_{\xi_i, \eta_j}$$

(where weights $w_i = w_j = 1$)

MATLAB Solver Architecture

1. Pre-Processing & Assembly

- **Mesh Parsing:**
  Custom function reads Gmsh ASCII `.msh` files to extract *Nodal Coordinates* and *Element Connectivity*.

- **Memory Allocation:**
  Global stiffness matrix (K) is initialized using MATLAB's `sparse()` function to optimize RAM.

- **Global Assembly:**
  Iterates through elements, computes local K_e, and maps it to global DOFs using scatter arrays.

2. Solution & Post-Processing

- **Boundary Conditions:**
  Essential (kinematic) constraints are enforced using the **Penalty Method** (adding 10^{15} to main diagonal).

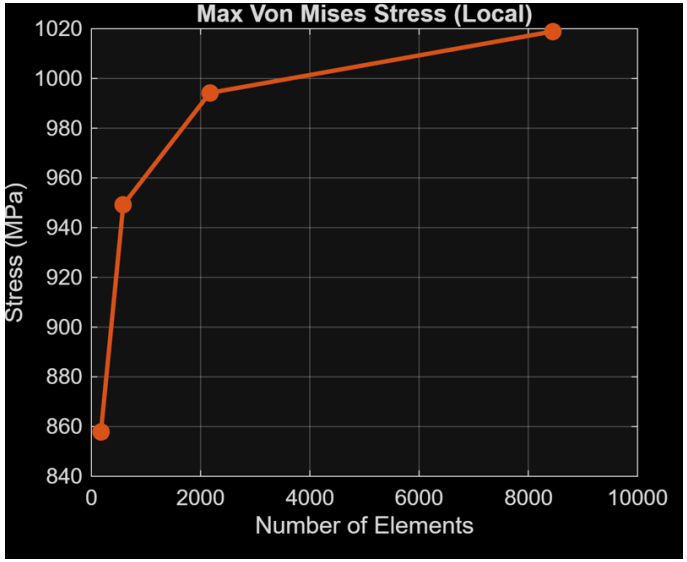
- **Linear Solver:**
  The massive system of equations $[KU = F]$ is evaluated via sparse matrix backslash factorization.

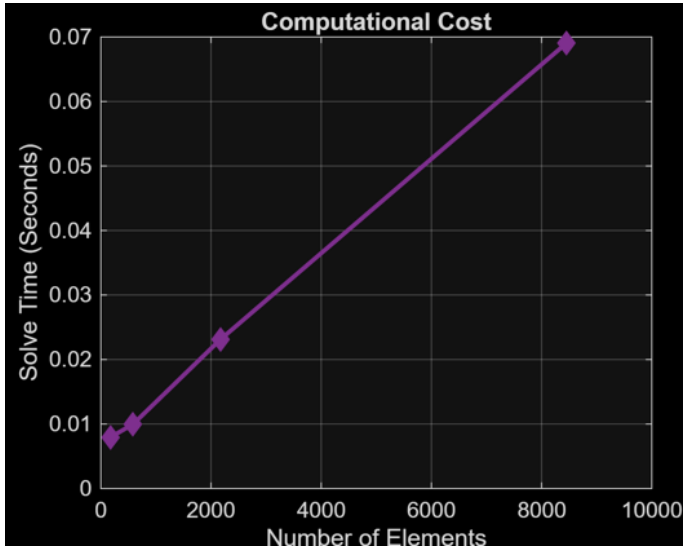
- **Stress Recovery:**
  Nodal displacements (U) are differentiated back into continuous Strain (ε) and Von Mises Stress fields.

Mesh Independence Study: Convergence vs. Cost

Objective: Balance discretization error ($h \to 0$) against computational cost.

- **Convergence:** Peak stress exhibits classic asymptotic behavior across the four meshes (Coarse to Ultra-Fine).
- **Cost Scaling:** Matrix inversion time increases exponentially with DOF count.
- **Verdict:** The Ultra-Fine mesh was selected for final analysis to guarantee absolute capture of the severe stress concentration gradients.

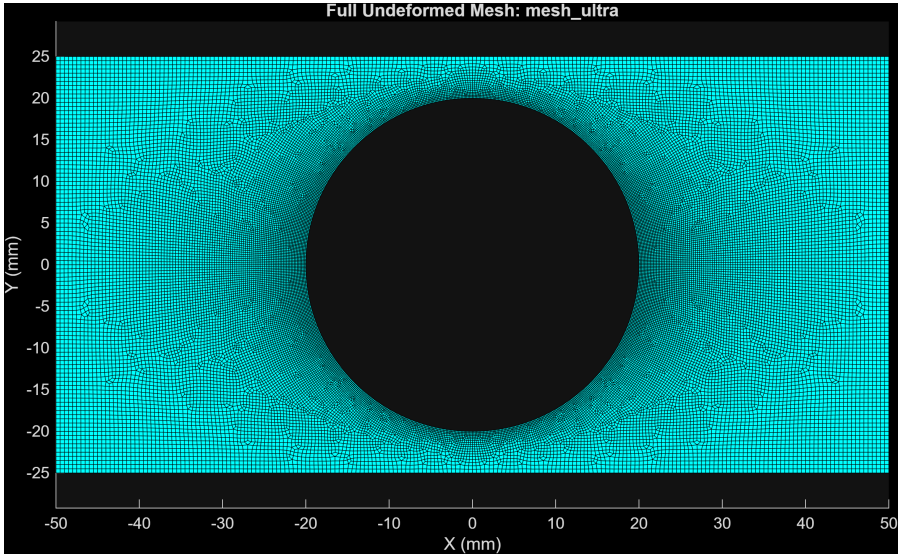




Results: Ultra-Fine Mesh & Kinematics

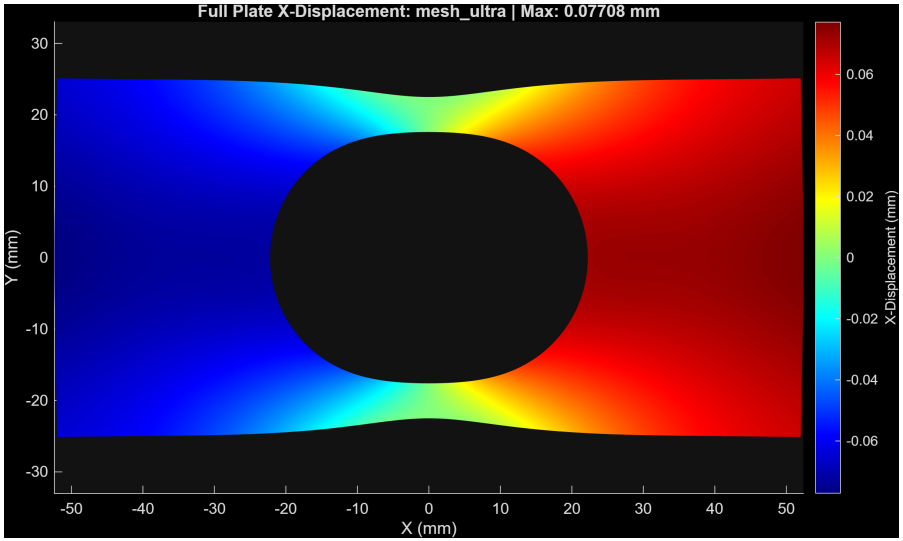
Full Plate Post-Processing: Quarter-symmetry arrays were mathematically reflected across the X and Y axes for full structural visualization.

Undeformed Mesh



Shows heavy local refinement at the discontinuity.

X-Displacement (U_x)

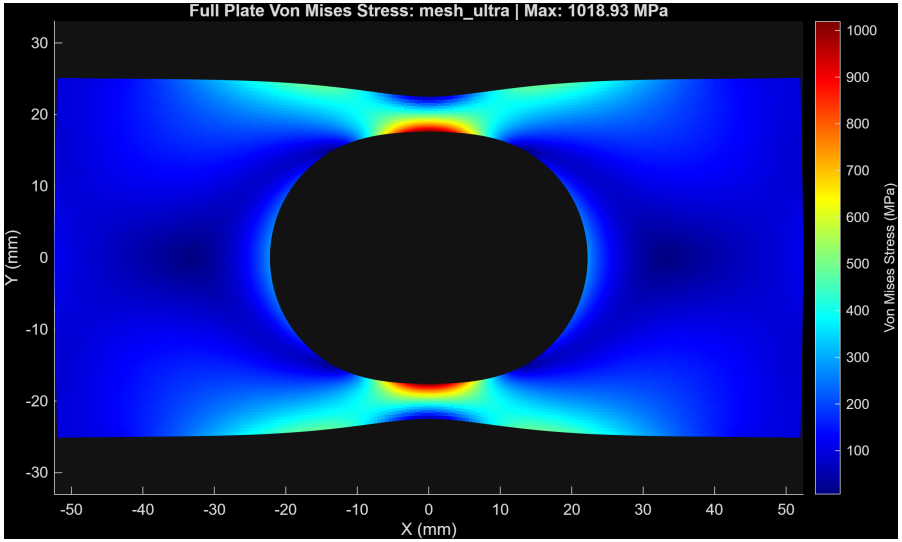


Symmetrical expansion under uniaxial tension.

Results: Stress & Strain Energy Density

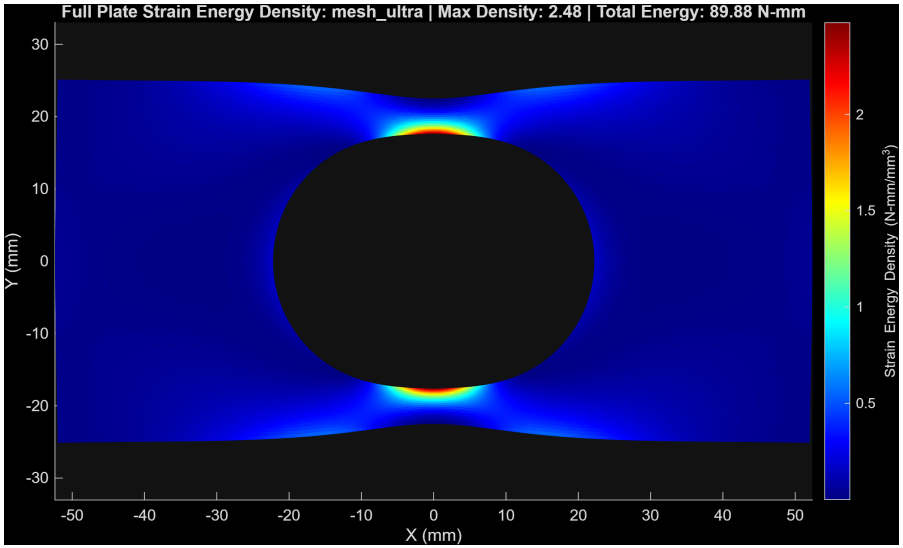
Concentration Capture: The high element density perfectly resolves the severe stress gradients along the Y-axis ligament.

Von Mises Stress



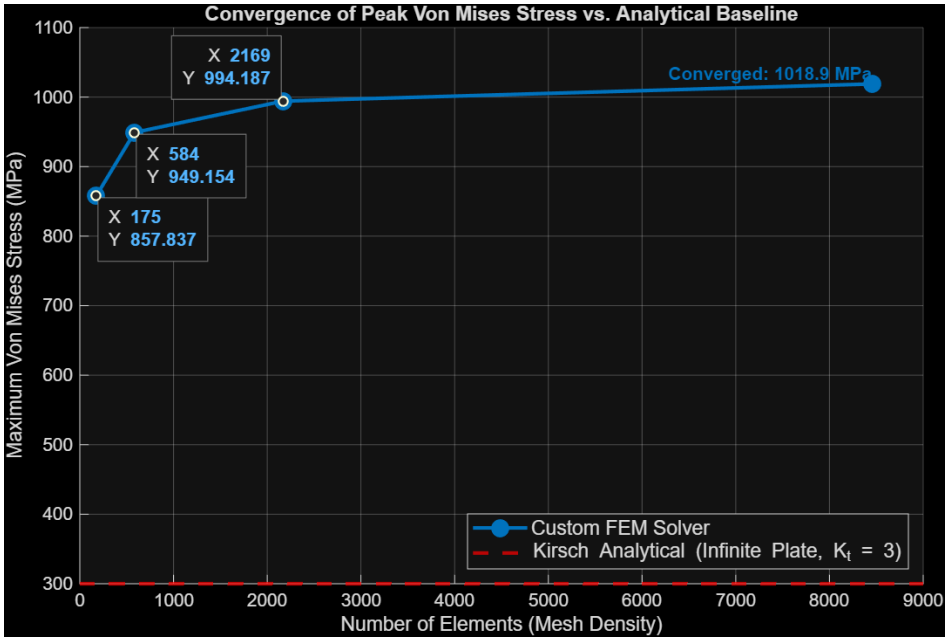
Peak stress highly localized at the apex of the void.

Strain Energy Density



Thermodynamic reflection of the stress concentration.

Analytical Validation with Kirsch's Theory



Infinite Plate Theory (Kirsch):

- Theoretical Concentration: $K_t = 3$
- Expected peak stress: $\sigma_{max} = 300$ MPa

Reason for Divergence:

- FEM solver converges to **1018.9 MPa**.
- **Why?** Kirsch assumes an infinitely wide plate with infinite room for stress lines to flow.
- Our physical model is highly constrained ($a/W = 0.8$).
- Forcing the load through a narrow 5 mm ligament massively amplifies stress well beyond the $K_t = 3$ assumption.

Mesh Strategy (ANSYS)

- Automatic mesh generation
- Refinement near hole (100 divisions), bottom edge (100 divisions with bias), left edge (30 divisions with bias), and upper edge (150 divisions).
- Uniform mesh division at the right edge.

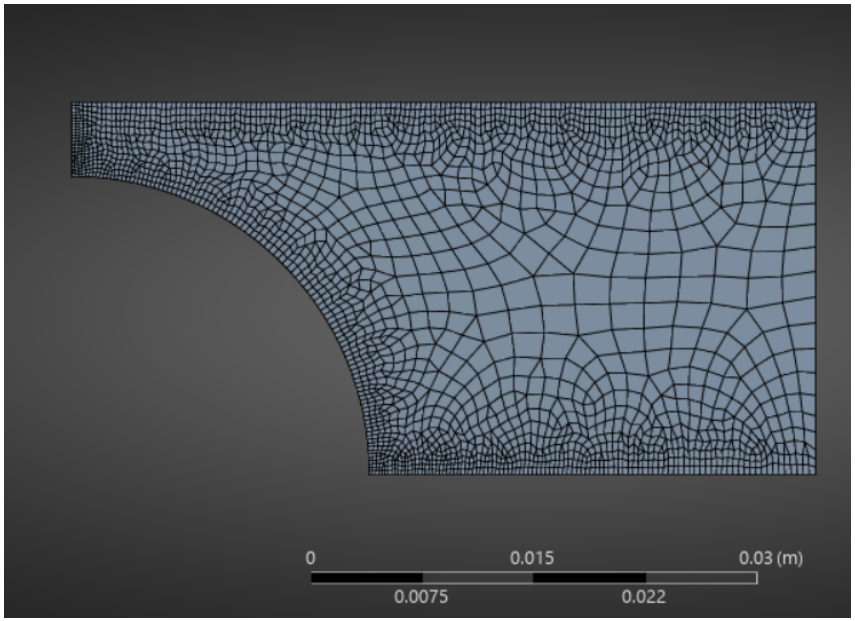


Figure: Q4 Mesh generated via Ansys

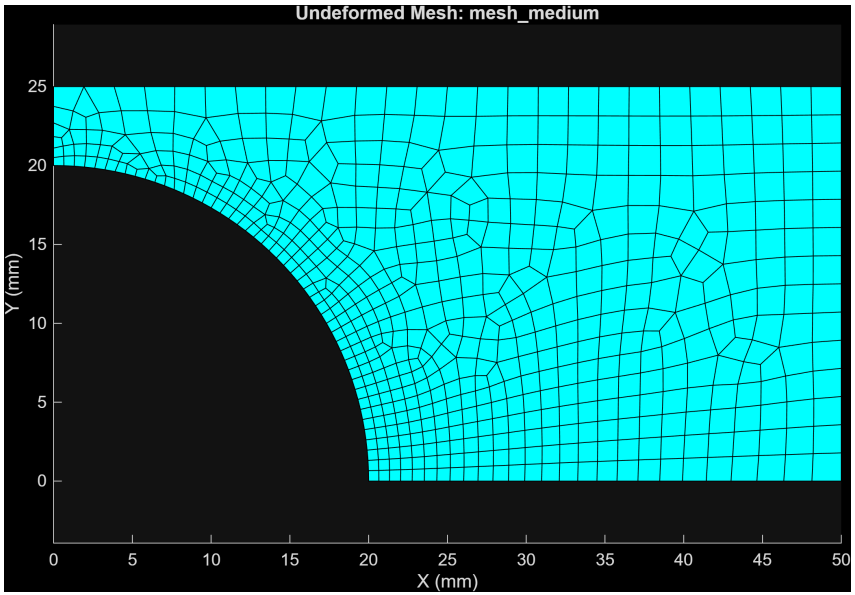
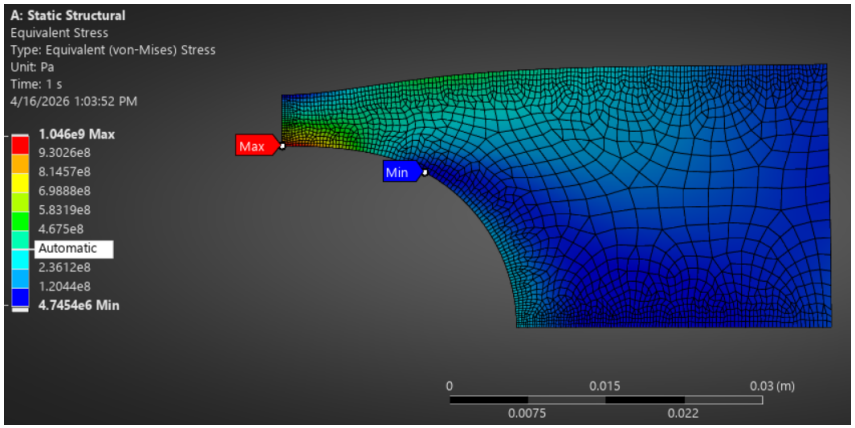


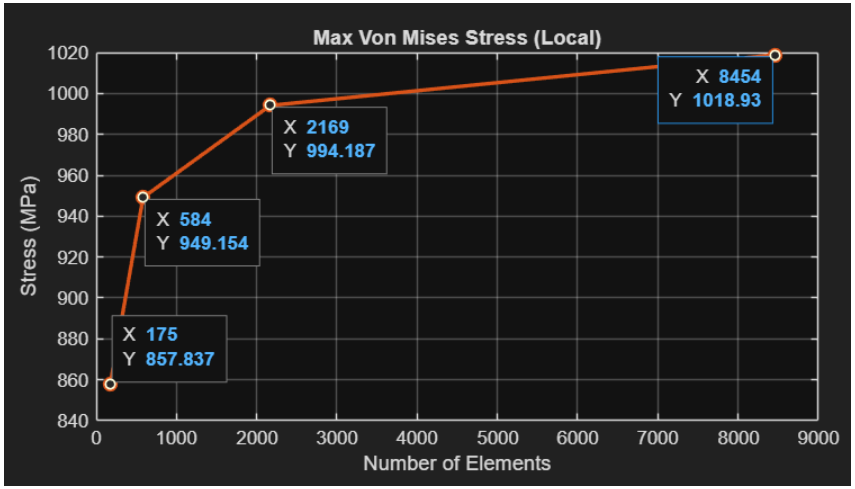
Figure: Q4 Mesh generated via Gmsh

Results (ANSYS)

- Equivalent (von-Mises) Stress



(a) Equivalent von-Mises stress



(b) Convergence of maximum stress

- ANSYS: 1.046 $\times 10^9$ Pa
- Gmsh/MATLAB: 1.018 $\times 10^9$ Pa
- Getting a difference of 2.7 %

Results (ANSYS)

- X-Displacement

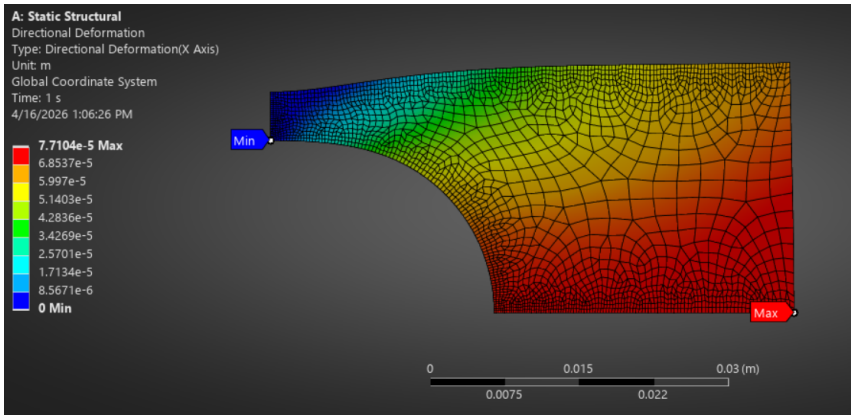


Figure: Directional deformation

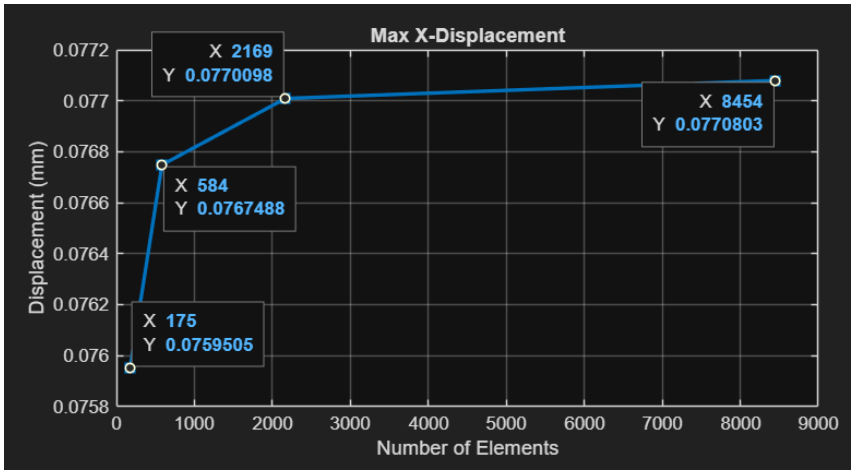


Figure: Convergence of Maximum X-Displacement

- ANSYS: 7.7104 $\times 10^{-5}$m
- Gmsh/MATLAB: 7.77 $\times 10^{-5}$ m
- Getting a difference of less than 1%.

Results (ANSYS)

- Strain Energy

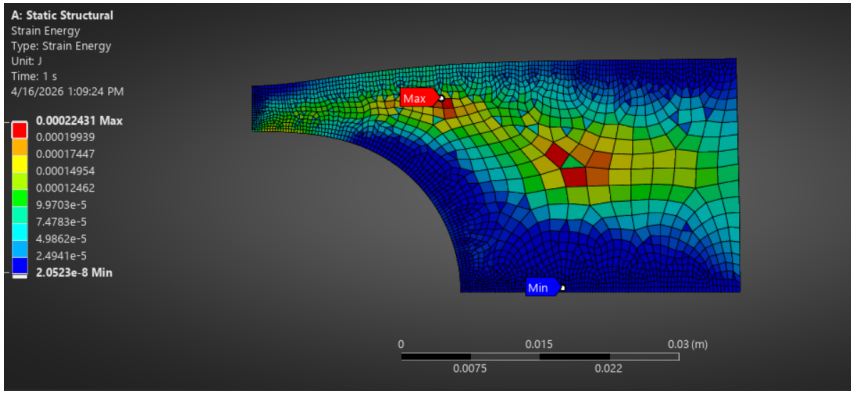


Figure: Strain Energy distribution

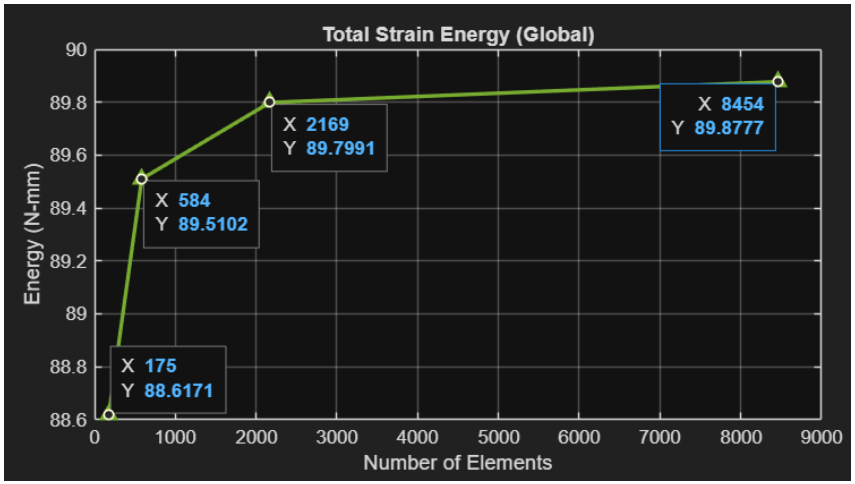


Figure: Energy convergence

- ANSYS: 2.2431 $\times 10^{-4}$ J
- Gmsh/MATLAB: 8.9877 $\times 10^{-2}$ J

Conclusion

- A custom 2D FEM solver for plane stress analysis was successfully developed and implemented in MATLAB.
- The solver accurately captured stress concentration effects in a plate with a circular hole.
- Strong agreement was observed between MATLAB and ANSYS results for:
  - Maximum von-Mises stress
  - X-direction displacement
- The computed stress concentration factor approached the theoretical value ($K_t \approx 3$), validating the physical correctness of the model.
- Minor discrepancies in strain energy highlight the sensitivity to implementation details such as the thickness assumptions.
- Overall, the project demonstrates a successful transition from theoretical FEM formulation to accurate computational implementation and validation.

Future Work

- Extend the solver to 3D problems and complex geometries.
- Implement higher-order elements (Q8, T6) for improved accuracy.
- Incorporate nonlinear material behavior and large deformation analysis.
- Automate mesh generation within MATLAB to remove dependency on external tools.
- Perform fatigue and fracture analysis near stress concentration regions.

References

- Plate with a hole analysis
- 2D plate 2D Plate with a Hole analysis using ANSYS Workbench
- ANSYS WorkBench — Plate with Hole

Thank You!